

Qazi Nasir (Ph.D.) Visa Status: Permeant Resident

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7515 Walton Lane Annandale, Virginia 22003

PROFESSIONAL HIGHLIGHTS

- ✓ **19+ Years of Experience** in Chemical & Petroleum Engineering, combining teaching, research, and industry expertise.
- ✓ **Secured Competitive Research Funding, leading projects** on gas hydrate modeling, CO₂ capture, catalytic hydrogenation, and AI-driven process optimization.
- ✓ **Published High-Impact Papers** in leading journals, covering, thermodynamics, catalysis, Gas Hydrate Prediction, Machine Learning, Nickel Doping, Catalyst Characterization, Nanotube Synthesis, Carbon Capture, Plasma Reactor, Hydrogen Production, Anaerobic Co-Digestion
- ✓ **Expert in Process Simulation & AI Applications**, skilled in Aspen Plus, HYSYS, MATLAB, Python, and Machine Learning for process optimization.
- ✓ **ABET-Accredited Curriculum Development, designing modern engineering courses** in chemical process simulation, thermodynamics, reservoir engineering, catalysis & catalytic reactions, gas hydrate, and natural gas processing
- ✓ **Leadership & Administration**, serving as **Acting Head of Department**, overseeing curriculum planning, faculty coordination, and strategic decision-making.
- ✓ **Industry & Academic Collaborations**, working with international teams on sustainable energy, CO₂ utilization, and process intensification.
- ✓ **Mentored Students**, supervising undergraduate and graduate research projects, guiding them toward successful academic and industrial careers.
- ✓ **Recognized Reviewer & Editorial Member**, contributing to leading journals like Elsevier, Springer, and Wiley in chemical and environmental engineering.
- ✓ **Hands-on Laboratory & Industrial Experience**, proficient in Gas Hydrate Studies, PVT analysis, Distillation, and Catalysis with real-world industrial applications and Ammonia Production, Sulfuric Acid (110 MTPD), and Granulated Single Super Phosphate (GSSP) (300 MTPD) plants

OBJECTIVE/ACADEMIC PROFILE

Chemical Engineering scholar with 19+ years of academic, research, and industry experience specializing in separations, extraction processes, thermodynamics, and CO₂ capture and utilization. Proven record of securing competitive international research funding (PETRONAS, TRC Oman, YUTP) and publishing extensively in high-impact journals. Skilled in scaling laboratory innovations to pilot systems, integrating AI and process modeling into chemical engineering solutions, and collaborating with industry partners on applied energy and resource challenges. Dedicated educator with extensive experience teaching core chemical and biomedical engineering courses, supervising graduate research, and contributing to ABET-accredited curriculum development. Seeking to advance the institute mission by developing innovative, externally funded research programs in critical, sustainable process design, and energy-resource integration, while fostering student success through impactful teaching and mentorship.

EDUCATION

Ph.D. in Chemical Engineering

2012 -2015

University Technology PETRONAS

Dissertation: Thermodynamic and Phase behaviour of fluid and clathrate hydrate of carbon dioxide rich gas mixture

MSc. In Petroleum Engineering

2008 - 2009

Teesside University

BSc. In Chemical Engineering

2000 - 2004

Institute of Chemical Engineering

PROFESSIONAL EXPERIENCE

ACADEMIC APPOINTMENTS/ TEACHING EXPERIENCE

Assistant Professor

2016 – 2/2025

Department of Chemical & Petrochemical Engineering, UNIZWA

Website: <https://www.unizwa.edu.om>

Lecturer / Assistant Professor

2010 - 2016

School of Chemical & Material Engineering (SCME), NUST

Website: <https://scme.nust.edu.pk/>

TEACHING HIGHLIGHTS

- Developed and taught core Chemical Engineering courses: Chemical Process Simulation, Chemical Engineering Thermodynamics, Plant Design I & II, Natural Gas Processing, Catalysis & Catalytic Reaction, Chemical Process Principles, Numerical Methods & Analysis, Fundamental of Chemical Engineering I & II, Natural Gas Processing and Gas Hydrate which equipping students with practical and theoretical knowledge critical for success in chemical and petrochemical engineering
- Designed flipped classroom modules and interactive simulation-based labs for Chemical Process Simulation and Thermodynamics courses, increasing student engagement by over 15%.
- Developed interdisciplinary teaching materials bridging chemical and biomedical engineering principles, preparing students for emerging research fields.
- Implemented project-based learning in Plant Design I & II, with several student projects recognized at departmental and national levels.
- Mentored over 50 undergraduate and graduate students, including those from underrepresented backgrounds, fostering inclusion and professional growth.
- Integrated AI and machine learning tools into senior design projects, equipping students with skills for Industry 4.0 and research applications.

CURRICULUM DEVELOPMENT & ACCREDITATION

- Led ABET-aligned curriculum redesign for multiple core chemical engineering courses, integrating industry needs and emerging research fields.
- Created new courses and modules in chemical process simulation, renewable energy systems, and biomedical process modeling.
- Designed laboratory modules for microfluidics, catalysis, and CO₂ capture, ensuring students gain hands-on, research-relevant skills.
- Served on ABET Accreditation Committee (2017–present), contributing to self-study reports, outcome assessments, and continuous improvement planning.

LABORATORY SAFETY EXPERIENCE

- Developed and implemented laboratory safety protocols for undergraduate teaching laboratories
- Conducted safety training for 100+ students in chemical engineering laboratories
- Ensured compliance with safety standards in research involving high-pressure equipment (PVT cells, gas chromatography)

STUDENT MENTORING & RECRUITMENT

- Mentored undergraduate and graduate students on academic research projects and career development
- Advised numerous final year projects, with 3 projects receiving departmental recognition
- Participated in university open houses and engineering career fairs to attract prospective students
- Developed departmental brochures and promotional materials for undergraduate recruitment
- Mentored PhD students in areas of hydrate inhibition, separation processes, and catalysis.
- Alumni placements include PhD programs, academic positions, and oil & gas industry roles.

DIVERSITY, EQUITY & INCLUSION COMMITMENT

- Mentored students from diverse cultural, academic, and socioeconomic backgrounds, promoting an inclusive and supportive learning environment.
- Integrated culturally responsive pedagogy into course delivery, ensuring accessibility and relevance for all learners.
- Actively engaged with initiatives to increase the participation of women and minorities in chemical and biomedical engineering.

RESEARCH HIGHLIGHTS

- Applied machine learning techniques to predict complex thermodynamic systems, with potential applications in biomedical process modeling
- Renewable Energies & Energy Storage: Developed catalytic hydrogenation of CO₂ to methanol for renewable fuel applications; investigated advanced catalytic systems for sustainable energy production.
- CO₂ Capture & Utilization: Led multiple projects on carbon capture, plasma-assisted CO₂ conversion, and integration into energy storage systems.
- Integrated AI and process optimization in chemical engineering, with transferable applications to biomedical systems
- Published 20+ peer-reviewed articles in high-impact journals, demonstrating research excellence

RESEARCH EXPERIENCE

Member of the research cluster

Application of non-thermal plasma in Science and Engineering

Department of Chemical & Petrochemical Engineering, UNIZWA

2016 – to

date

- Conducted advanced thermodynamic studies, developed mathematical models, and applied machine learning techniques to analyze vapor-liquid equilibria and gas hydrate systems in PVT Cell.
- Published extensively in high-impact, peer-reviewed journals, contributing to the academic and industrial knowledge base.
- Secured competitive funding for cutting-edge energy research projects, demonstrating strong grant-writing and project management skills.
- Collaborated with interdisciplinary teams to innovate plasma-enhanced gas processing techniques, bridging expertise across fields to achieve project objectives.

Research Center for CO₂ Capture (RCCO₂C), UTPWebsite: <https://bit.ly/45GYN5Z>

- Conducted in-depth thermodynamic and phase behavior studies of vapor-liquid equilibria and gas hydrates, using advanced mathematical modeling and lab measurements (PVT Cell).
- Published multiple research papers, achieving all project milestones, and generated quarterly reports that contributed to project success.
- Established key research alliances and collaborative agreements with external organizations, resulting in enhanced project scope and industry partnerships.
- Led the coordination and successful execution of project meetings, workshops, and technical events, improving interdisciplinary collaboration.

RESEARCH FUNDING & GRANTS

- Propane oxidative dehydrogenation using catalytic membrane reactor **2023**
Role: Investigator
Funding body: The Research Council (TRC), Grant # BFP/RGP/EI/18/023
Duration: 12 months
- Kinetic study of hydrate formation: Flow assurance and separation of CO₂ rich gas mixture **2012 - 2015**
Role: Investigator
Funding body: PETRONAS Research SDN BHD (PRSB).
Duration: 36 months
- Lab scale measurements and mathematical modeling of the phase behavior of CO₂ rich gas mixture with conditions (pressure & temperature) replicating reservoir to surface conditions **2012 - 2015**
Role: Investigator
Funding body: Yayasan Universiti Teknologi PETRONAS- Fundamental Research (YUTP) fund
Duration: 36 months

RESEARCH INTEREST

- **PVT Gas Hydrate Study:** Investigating the pressure-volume-temperature behavior of gas hydrates to understand their formation and decomposition mechanisms.
- **Thermodynamic and Phase Behavior Prediction:** Utilizing advanced mathematical models to predict thermodynamic and phase behavior in complex systems.
- **Development of Hydrate Inhibitors and Promoters:** Innovating and optimizing hydrate inhibitors and promoters to address flow assurance challenges and enhance storage solutions.
- **CO₂ Capture and Utilization:** Exploring efficient methods for capturing carbon dioxide and converting it into valuable products, contributing to sustainable environmental practices.
- **Catalytic Hydrogenation of Carbon Dioxide to Methanol:** Developing and optimizing catalytic processes for converting carbon dioxide into methanol, a valuable chemical feedstock.
- **Artificial Intelligence in Chemical Engineering:** Leveraging AI for improved process optimization, intelligent materials design, advanced process control, and predictive modeling, aiming to revolutionize chemical engineering practices through data-driven approaches.

INDUSTRIAL EXPERIENCE**Apprentice/ Process Engineer****2004 - 2008**

Pak-American Fertilizer PVT (Ltd)

Website: <https://bit.ly/3KXAcSv>

- Supervised the water treatment plant and laboratory operations to ensure high production quality and compliance with stringent standards.

- Conducted comprehensive laboratory analyses of raw materials, finished products, and chemicals to maintain optimal process outcomes.
- Led a multidisciplinary team as an Independent Shift Engineer, ensuring smooth plant operations and effective emergency management.
- Monitored and optimized critical process parameters to enhance operational efficiency and productivity.
- Utilized advanced process modeling tools, including Aspen HYSYS, Aspen Plus, ChemStation, CHEMCAD, and ProSim, to improve process performance and achieve high-accuracy simulations.

RESEARCH PUBLICATIONS

BOOK CHAPTER

1. Hydrate formation challenges in the pipelines, Adv. In G. Gases Vol. 3- Greenhouse Gases Storage and Transportation, Elsevier

PAPERBACK BOOKS (AMAZON)

1. Qazi Nasir, Thermodynamic and Phase behavior study of Fluid and Gas Hydrate (2017), <https://amzn.to/3ONVXXy>
2. Qazi Nasir, Transforming Business Operations for Improved Efficiency and Profitability: The Role of Artificial Intelligence in Manufacturing, <https://bit.ly/43c4rMy>
3. Qazi Nasir, Fueling the Future: The Revolutionary Guide to Biodiesel Production, <https://bit.ly/43DygoX>
4. Qazi Nasir, Exploring the Potential of Gas Hydrates: A Comprehensive Guide to Applications, <https://bit.ly/3MNKu7M>

JOURNAL PUBLICATIONS (SELECTED)

1. Israf Ud Din, Jamshid Khan, Qazi Nasir (2025), The Opportunities and Challenges of Using Nanomaterials as Photocatalysts for the Degradation of Environmental Pollutants, Comments on Inorganic Chemistry 1 (46), <https://doi.org/10.1080/02603594.2025.2451829>
2. H Gul, M Bilal, T Hussain, IU Din, M Mubashir, Qazi Nasir (2024), Effects of Ni Loadings on the Structure and Morphology of Carbon Nanotubes Using Nickel Doped Iron Oxide Catalysts, ChemistrySelect 9 (38), <https://doi.org/10.1002/slct.202402285>
3. G Ngwu, H Suleman, F Ahmad, D Qadir, Z Shamair, Qazi Nasir, M Nawaz (2024), Passive Direct Air Capture of Carbon Dioxide with an Alkaline Amino Acid Salt in Water-based Paints, Energies, <https://doi.org/10.3390/en17020320>
4. Qazi Nasir, H Suleman, WS Abdul Majeed (2023), Application of machine learning on hydrate formation prediction of pure components with water and inhibitors solution. The Canadian Journal of Chemical Engineering, <https://doi.org/10.1002/cjce.25291>
5. Wameath S Abdul Majeed, Qazi Nasir, Muzna H Alajmi, Khaloud A Almaqbali, (2023), Evaluation of two plumes jet plasma reactor for plasmolysis of H₂O vapor and CO₂ combination-optimization study, Journal of Advanced Manufacturing and Processing, <https://doi.org/10.1002/amp2.10154>
6. Anwar Ahmad, Roomana Ghufuran, Qazi Nasir, Fathima Shahitha, Mohammed Al-Sibani, Amal S. Alpredicting -Rahbi, (2023), Enhanced anaerobic co-digestion of food waste and solid poultry slaughterhouse waste using fixed bed digester: Performance and energy recovery, Environmental Technology & Innovation, <https://doi.org/10.1016/j.eti.2023.103099>
7. Qazi Nasir, Wameath S Abdul Majeed, and Humbul Suleman, (2023), Predicting Gas Hydrate Equilibria in Multicomponent Systems with a Machine Learning Approach. Chem. Eng. Technol. <https://doi.org/10.1002/ceat.202300055>
8. Bruno Imai, Qazi Nasir, Abdulhalim Shah Maulud, Muhammad Nawaz, Rizwan Nasir & Humbul Suleman (2023), Neural network-based correlation and statistical identification of data outliers in H₂S-alkanolamine-

H₂O and CO₂-alkanolamine-H₂O datasets, Neural Computing and Applications, <https://doi.org/10.1007/s00521-022-07904-z>

9. Israf Ud Din*, Qazi Nasir, Mustapha D. Garba, Abdulrahman I. Alharthi, Mshari A. Alotaibi and Muhammad Usman, (2022), A Review of Preparation Methods for Heterogeneous Catalysts, Mini-Reviews in Organic Chemistry, DOI: 10.2174/1570193X18666210308151136
10. Qazi Nasir, Humbul Suleman, Israf Ud Din & Yasir Elsheikh Elfadol , (2022), A multi-layer perceptron neural network model for predicting the hydrate equilibrium conditions in multi-component hydrocarbon systems, Neural Computing and Applications, <https://doi.org/10.1007/s00521-022-07284-4>
11. Yasir A. Elsheikh, Wameath S. Abdul-Majeed, Qazi Nasir, In situ Transesterification/Reactive Extraction of Castor Bean Seeds Assisted by Flying Jet Plasma for Biodiesel Production, Appl. Chemical. Eng, <https://doi.org/10.14478/ace.2022.1086>
12. Humbul Suleman, Abdulhalim Shah Maulud, Philip Loldrup Fosbøl, Qazi Nasir, Rizwan Nasir, Muhammad Zubair Shahid, Muhammad Nawaz, Mustafa Abunowara, (2021), A review of semi-empirical equilibrium models for CO₂-alkanolamine-H₂O solutions and their mixtures at high pressure, Journal of Environmental Chemical Engineering, <https://doi.org/10.1016/j.jece.2020.104713>
13. Yasir A. Elsheikh, Fawzi Elfghi, Qazi Nasir, Nawshad Muhammad, (2020), Effect of pyrazolium ionic liquid halide content on in-situ transesterification of Castor Bean (*Ricinus Communis* L.) seeds, Process Safety and Environmental Protection, <https://doi.org/10.1016/j.psep.2020.05.014>
14. Israf Ud Din, Maizatul S. Shaharun, Abdul Naeem, Mshari A. Alotaibi, Abdulrahman I. Alharthi ,Qazi Nasir, (2020), CO₂ Conversion to Methanol over Novel Carbon Nanofiber-Based Cu/ZrO₂ Catalysts—A Kinetics Study, Catalyst, <https://doi.org/10.3390/catal10050567>
15. Qazi Nasir, Humbul Suleman, Yasir A. Elsheikh, (2020), A review on the role and impact of various additives as promoters/ inhibitors for gas hydrate formation, Journal of Natural Gas Science and Engineering, <https://doi.org/10.1016/j.jngse.2020.103211>
16. Qazi Nasir, Humbul Suleman, and Manivannan Elumalai, (2020), A Hybrid Membrane-Absorption Process for Carbon Dioxide Capture: Effect of Different Alkanolamine on Energy Consumption, Iran. J. Chem. Eng, <https://doi.org/10.30492/ijcce.2019.35853>
17. Israf Ud Din, Maizatul S. Shaharun, A. Naeem, Mshari A. Alotaibi, Abdulrahman I. Alharthi, M. Afroz Bakht, Qazi Nasir, (2020), Carbon nanofibers as potential materials for catalysts support, a mini-review on recent advances and future perspective, Ceramics International, <https://doi.org/10.1016/j.ceramint.2020.04.275>
18. Israf Ud Din; Maizatul S. Shaharun; A. Naeem; Mshari A. Alotaibi; Abdulrahman I. Alharthi; Qazi Nasir, (2020), Effect of reaction conditions on the activity of novel carbon nanofiber-based Cu/ZrO₂ catalysts for CO₂ hydrogenation to methanol. Comptes Rendus. Chimie, <https://doi.org/10.5802/crchim.6>
19. An Overview of Cryogenic Separation Techniques for Natural Gas with High CO₂ Content, Bhajan Lal, Azmi Mohd Shariff, Hilmi Mukhtar, Qazi Nasir and Ali Qasim, (2018), Journal of Engineering and Applied Sciences, <http://dx.doi.org/10.3923/jeasci.2018.2152.2155>
20. Qazi Nasir, Bhajan Lal, Lau Kok Keong, (2016), The Presence of Additive Mixtures (THF/SDS) on Carbon Dioxide Rich Gas Mixture Hydrate Formation and Dissociation Conditions, Advanced Materials Research, <https://doi.org/10.4028/www.scientific.net/AMR.1133.639>
21. Qazi Nasir, Khalik M. Sabil, K.K. Lau, (2015), Measurement of isothermal (vapor + liquid) equilibria, (VLE) for binary (CH₄+CO₂) from T = (240.35 to 293.15) K and CO₂ rich synthetic natural gas systems from T = (248.15 to 279.15) K, Journal of Natural Gas Science and Engineering, <https://doi.org/10.1016/j.jngse.2015.08.045>.
22. Israf Ud Din, S. Tasleem, A. Naeem, Maizatul S. Shaharun & Qazi Nasir, (2016), Study of Annealing Conditions on Particle Size of Nickel Ferrite Nanoparticles Synthesized by Wet Chemical Route, Synthesis and Reactivity in Inorganic, Metal-Organic, and Nano-Metal Chemistry, DOI: 10.1080/15533174.2014.988226
23. Humbul Suleman, Qazi Nasir, Maulud, A.S. Man, Z, (2015), Comparative Study of Electrolyte Thermodynamic Models for Carbon Dioxide Solubility in Water at High Pressure, Chemical Engineering Transactions, <https://doi.org/10.3303/CET1545099>

24. Qazi Nasir, K. K. Lau, Bhajan Lal, and Khalik M. Sabil, (2014), Hydrate Dissociation Condition Measurement of CO₂-Rich Mixed Gas in the Presence of Methanol/Ethylene Glycol and Mixed Methanol/Ethylene Glycol + Electrolyte Aqueous Solution, Journal of Chemical Engineering Data, <https://doi.org/10.1021/je5008313>
25. Khalik M. Sabil, Qazi Nasir, Bezhad Partoon, and Akbar A. Seman, (2014), Measurement of H-LW-V and Dissociation Enthalpy of Carbon Dioxide Rich Synthetic Natural Gas Mixtures, Journal of Chemical Engineering Data, <https://doi.org/10.1021/je500453j>
26. Qazi Nasir, K.M. Sabil and Khashayar Nasrifar, (2014), Measurement and Phase Behavior Modeling (Dew Point + Bubble Point) of CO₂ Rich Gas Mixture, Journal of Applied Sciences, <https://doi.org/10.3923/jas.2014.1061.1066>
27. Qazi Nasir, Kok Keong lau, Bhajan Lal, Enthalpies Of Dissociation Of Pure Methane And Carbon Dioxide Gas Hydrate, (2014). Zenodo, <https://dx.doi.org/10.5281/zenodo.1095928>
28. Qazi Nasir, K.M. Sabil and Khashayar Nasrifar (2014), CO₂ Rich Gas Mixture Hydrate-Liquid Water-Vapor (H-Lw-V) Equilibrium Measurement and Prediction, Applied Mechanics and Materials, <https://doi.org/10.4028/www.scientific.net/AMM.625.386>

CONFERENCE PROCEEDINGS

1. Yasir A. Elsheikh, Al-Hasnaa Al-Hoqani, Qazi Nasir, "In-situ transesterification of Ricinus communis seeds for biodiesel production", 8th National Symposium on Engineering Final Year Project, College of Engineering, University of Nizwa, May 2nd, 2018.
2. Qazi Nasir, Khaloud Aamir, "Recovery of Natural Gas through Hybrid Scheme of Membrane and Amine Unit", 8th National Symposium on Engineering Final Year Project, College of Engineering, University of Nizwa, May 2nd, 2018.
3. Humbal Suleman, Qazi Nasir, Abdul halim Shah Maulud, Zakaria Man "Study of electrolyte thermodynamic models for carbon dioxide solubility in water at high pressure", 18th conference process integration, modeling and optimization for energy saving and pollution reduction, Kuching, Sarawak, Malaysia, 2015.
4. Qazi Nasir, Lau Kok Keong, K. M. Sabil, Bhajan Lal, "CO₂ rich gas mixture hydrate-liquid water-vapor (H-Lw-V) equilibrium measurement & prediction using thermodynamic mathematical modeling", The 3rd International conference on process engineering and advanced materials, Kuala Lumpur, Malaysia, 2014.
5. Qazi Nasir, K. M. Sabil, K. nasrifar, "Measurement and phase behavior modeling (dew point + bubble point) of CO₂ rich gas mixture", International oil and gas symposium and exhibition (IOGSE), Sabah, Malaysia, 2013.
6. Qazi Nasir, K. M. Sabil, "Experimental measurement and prediction of hydrate equilibrium conditions for CO₂ rich gas mixture", 6th engineering conference on environment and energy (EnCon), Kuching Sarawak, Malaysia, 2013.

PROFESSIONAL SERVICE & STEM OUTREACH HIGHLIGHTS

- Reviewer for leading journals (Elsevier, Springer, Wiley) and funding agencies, including The Research Council.
- Organized technical workshops and faculty development programs in renewable energy, chemical process simulation, and research methodology.
- Participated in STEM outreach programs for high school students, encouraging participation from underrepresented communities in engineering.
- Contributed to departmental recruitment through open houses, engineering fairs, and community engagement events.
- Collaborated with industry partners (PETRONAS, Oman's TRC, fertilizer industry) on applied separation and process engineering projects.

- Member of ABET Accreditation Committee — contributed to curriculum alignment and program assessment.
- Organized and participated in international workshops on energy, CO₂ capture, and separation technologies

ACADEMIC

- ✍ Internal and external evaluators of postgraduate thesis and research synopsis.
- ✍ Evaluate internship reports, final year design and chemical process design projects at the undergraduate

MEMBERSHIPS

- Pakistan Engineering Council
- Institute of Chemical Engineers (IChemE)
- International association of carbon capture
- International association of Engineers
- International Statistical Engineering Association

REVIEWER ASSIGNMENTS

- ✍ Journal of Open Innovation: Technology, Market, and Complexity (Elsevier)
- ✍ Carbon Capture Science & Technology (Elsevier)
- ✍ Biomass conversion and biorefinery (Springer Nature)
- ✍ Environment technology & innovation (Elsevier)
- ✍ Chemical Engineering Science (Elsevier)
- ✍ Journal of open innovation (Elsevier)
- ✍ Journal of petroleum exploration and production technology (Springer Nature)
- ✍ Case studies in Chemical and Environment Engineering (Elsevier)
- ✍ Chemical Engineering Science (Elsevier)
- ✍ Ministry of higher education, research and innovation – TRC (Project reviewed)

AWARDS & PROFESSIONAL DEVELOPMENT (TRAINING & WORKSHOP)

- ☆ Graduate Assistant Scholarship for PhD studies (UTP)
- ☆ National Fertilizer Cooperation Scholarship during undergraduate (NFC)
- ☆ Secure Second (2nd) Position in College during Higher Secondary School Examination
- 📖 Research Methodology Course (UTP)
- 📖 Attended workshop on Publication in first-class scientific journals: strategy and steps (UTP)
- 📖 Attended the short course on Oil and Gas Projects Management, conducted by TOTAL. Certificates were provided by the Total Professor Association (TPA)
- 📖 Certificate of Completion Hands-On Introduction: React (LinkedIn)
- 📖 Certificate of Completion Python Programming Efficiently (LinkedIn)
- 📖 Certificate of Completion Double Down Bet on Yourself and Succeed on your Own Terms (LinkedIn)
- 📖 Certificate of Completion How to Develop your Career Plan (LinkedIn)
- 📖 Certificate of Completion Machine Learning with Python Foundations (LinkedIn)
- 📖 Certificate of Completion Python Functions for Data Science (LinkedIn)
- 📖 Certificate of Completion Machine Learning with Scikit Learn (LinkedIn)
- 📖 Certificate of Completion Python for Engineers and Scientists (LinkedIn)
- 📖 Certificate of Completion Python Data Structures Sets and Frozen Sets (LinkedIn)
- 📖 Certificate of Completion Learning VBA in Excel (LinkedIn)
- 📖 Certificate of Completion: How to Research and Writing using Generative AI Tools (LinkedIn)

- 📄 Certificate of Completion Python for Nonprogrammers (LinkedIn)
- 📄 Urban Flood Management and Flood Risk Mapping (PEC)
- 📄 Solar and Wind Power Potential in Pakistan (PEC)
- 📄 Certification of Completion Deep Learning onramp (MathWorks)
- 📄 Certificate of Completion Machine Learning Onramp (MathWorks)

TEACHING-RELATED PROFESSIONAL DEVELOPMENT

- 📄 Certificate of Completion: How to Research and Write using Generative AI Tools (LinkedIn)
- 📄 Certificate of Completion: Machine Learning with Python Foundations (LinkedIn)
- 📄 Attended workshop on Publication in first-class scientific journals: strategy and steps
- 📄 Participated in faculty development programs on innovative teaching methodologies

HARD SKILLS

TECHNICAL SKILLS

- 💻 Chemical Process Simulation software's: Aspen HYSYS, Aspen Plus, Aspen Custom Modular, ProSim, ChemCAD, COMSOL Multiphysics, OpenLCA, Minitab, Multiflash, AutoCAD

PROGRAMMING AND SOFTWARE DEVELOPMENT SKILLS

- 💻 MathWorks MATLAB, Python, Visual Basic, WordPress, React.js, R-language

MACHINE LEARNING AND DATA ANALYSIS

- 💻 Strong problem-solving abilities, Machine learning algorithms (supervised and unsupervised), Data preprocessing, Model development, Feature Engineering, Model tuning, Deep learning frameworks, data visualization, Kaggle, Google Colab & Jupyter notebook, Pandas, NumPy, sklearn, SciPy, openpyxl, selenium

OFFICE APPLICATION

- 💻 Microsoft Office Suite: MS Word, MS Excel, & MS PowerPoint, Endnote, Mendeley

LABORATORY EQUIPMENT AND INSTRUMENTS EXPERIENCE:

- 🔧 Vinci PVT system, Gas hydrate Cell, Gas Chromatography (GC), High pressure liquid chromatography (HPLC), pilot scale distillation unit, Fixed bed absorption column.

SOFT SKILLS

Category	Description
Effective Communication	Proficient in articulating complex research concepts clearly and concisely in peer-reviewed papers, grant proposals, and technical reports. Skilled in delivering impactful research presentations at academic conferences, seminars, and to interdisciplinary teams.
Team Collaboration	Experienced in collaborating within diverse, interdisciplinary research teams to achieve synergistic outcomes in complex projects. Active contributor to academic committees, facilitating collective decision-making and advancing departmental research goals.
Leadership	Proven leadership as a principal investigator and mentor, guiding graduate and undergraduate researchers toward successful project milestones. Facilitated research collaborations with external institutions and industry partners to foster innovation and expand research impact.

Problem Solving	Demonstrated advanced problem-solving skills by devising innovative solutions to address challenges in experimental design, data analysis, and process optimization. Applied strong analytical skills to tackle complex research questions and achieve breakthrough results in the field.
Adaptability	Adapted seamlessly to diverse research environments and methodologies through international collaborations and interdisciplinary projects. Proactively addressed evolving research challenges, ensuring project continuity and resilience under varying conditions.
Time Management	Effectively managed time to balance extensive research projects, grant writing, academic publishing, and mentoring responsibilities. Consistently met tight project deadlines, optimized laboratory workflows, and maintained a productive research output.

WEB METRIC

Website:	https://qazinasir.com
Research:	https://research.qazinasir.com
Blogs:	https://blog.qazinasir.com
Research Gate:	https://goo.gl/lZgDEm
ORCID:	https://orcid.org/0000-0002-5642-7957
Google Scholar:	https://goo.gl/uYPMYF
Academia:	https://bit.ly/3vnvoz1
Scopus	https://bit.ly/3WgsaK7

